

Aliah University

End Semester Examination, Jan. 2022

Course: B. Tech. (Civil Engineering)

Year & Semester: 3rd Yr. (V Sem.)

Subject: Design of Steel Structures

Code: CENUGPC12

Full Marks: 80

Time: 3hrs.

Answer any four questions. Assume suitable data if required.

1. (i) What do you mean by shape factor? Explain. Calculate the shape factor of a rectangular cross section.

(ii) Determine the size and effective length of the side fillets to connect two plates of 120×10 mm and 100×10 mm which are subjected to a factored tension of $(140+X)$ kN, where X= last two digit of your Roll No. The ultimate strength of the plates, $f_u = 410$ MPa.

[10+10=20]

2. Design a bridge truss diagonal of 2.6 m length subjected to a factored tensile load of $(220+X)$ kN, where X= last two digit of your Roll No. The tension member is connected to a gusset plate 16 mm thick with one line of 20 mm diameter bolts of grade 8.8. (Given $f_y = 250$ MPa, $f_u = 410$ MPa, $f_{ub} = 800$ MPa and $A_{nb} = 245$ mm²)

[20]

3. Design a single equal angle to carry a compression of $(60+X)$ kN, where X= last two digit of your Roll No. The centre to centre distance between the end connections is 2.0 m. Assume that the end connection is done by at least two bolts. Design the end connection also. The grade of steel is E250.

[20]

4. (i) Explain the classification of sections and their basis of classification as described in IS 800:2007

(ii) A simply supported beam of span 6 m carrying a distributed load of 30 kN/m (including self-weight) throughout the span. Design the beam assuming that it is laterally supported along its length. The grade of steel is E250.

[8+12=20]

5. Calculate the design load carrying capacity in compression which is 2.5 m long. (the centre to centre distance between the end connections) for the following sections. (i) ISMB 200, (ii) ISMC 200 and (iii) Two angles $75 \times 75 \times 8$ connected to both sides of 10 mm thick gusset by two bolts which is a discontinuous strut. The grade of steel is E250.

[6+6+8=20]

ALIAH UNIVERSITY

(Dept. of Civil Engg.)

End-Semester Examination- January, 2022

Year: 3rd Year

Subject: Open Channel Flow

Full Marks: 80

Semester: 5th Semester

Subject Code: CENUGPC13

Time: 3 hr.

Attempt all question and assume suitable data if required.

1. (a) Discuss the following: 8
 - (i) Open channel flow and pipe flow.
 - (ii) Prismatic, non prismatic channel, rigid and mobile boundary channel.
 - (iii) Steady-unsteady flow, Uniform-non uniform flow, gradually and rapidly flows.
 - (iv) Velocity distribution and pressure distribution on linear and curvilinear flow.

(b) The velocity distribution in a channel section may be approximated by the equation, $V = V_o(y/y_o)^n$, in which V is the flow velocity at depth y ; V_o is the flow velocity at depth y_o , and n = a constant. Derive expressions for the energy and momentum coefficients. 8
2. (a) A 5.0 m wide rectangular channel carries a discharge of 20 m³/s at a depth of 2.0 m. The width beyond a certain section is to be changed to 3.5 m. If it is desired to keep the water surface elevation unaffected by this change, what modification is needed to the bottom elevation? 6

(b) A triangular channel has an apex angle 60° and carries a flow with a velocity of 2 m/s and depth of 1.25 m. (i) What is the critical depth? (ii) What is specific energy? 4

(c) A rectangular channel is 4.0 wide and carries a discharge of 20 m³/s at a depth of 2.0 m. At a certain section it is proposed to build a hump. Calculate the water surface elevation at the upstream of hump and over the hump if the hump height is 0.33 m. 6
3. (a) A 3.6 m wide rectangular channel had a manning's $n=0.03$ and its bed was lined with concrete ($n= 0.015$). If the depth remains same before and after the repair, what is the increase of discharge obtained as a result of repair? 4

(b) What do you mean by equivalent roughness? Derive mathematical expression for equivalent roughness using Horton's Method. 4

(c) A trapezoidal channel with one side vertical the other sloping at 2H:1V carries a discharge of 28 m³/s. Determine the longitudinal slope and cross sectional dimension for best hydraulic efficiency if Manning's $n=0.014$. 8
4. (a) Discuss gradually varied flow with suitable assumption and mathematical expression. 5

(b) Discuss gradually varied flow classification and feature of flow profile 5

(c) In a 4 m wide rectangular channel ($n=0.017$) the bed slope is 0.0006. Channel is conveying 10.0 m³/s of flow, estimate nature of GVF profile for channel depth as 1.6 m and 2.1 m respectively. 6
5. (a) Discuss the various types of jump and its characteristics. 8

(a) In a hydraulic jump taking place in a rectangular channel the discharge intensity and head loss are found to be 4.7 m³/s/m and 6.0 m respectively. Determine the sequent depths of the jump. 8

Aliah University
End Term Examination (Odd Semester), 2021-22

Course: Civil Engineering; Year: 3rd Year; Subject: Transportation Engg.
Subject Code: CENUGPC15 Max. Marks: 80 Times: 3 hrs

Answer all questions

Q.1 a) Write short notes on: (i) Scope of Highway Engineering (ii) Factors Controlling Alignment.

b) What are the various surveys to be carried out before planning a highway system for a given area? Explain briefly.

(12+8)

Q. 2. a) What are the conditions which necessitate taking up of a re-alignment project of the highway?

b) Explain briefly the various stages of work in a new Highway Project.

(10+10)

Q.3. a) Derive an expression for calculating overtaking sight distance on a highway.

b) Find the stopping sight distance for a design speed of 65 kmph. Assume suitable data. What are the sight distance requirements at a gradient of 1 in 40?

(8+12)

Q.4 Explain superelevation. What are the factors on which the design of superelevation depends? Explain maximum and minimum superelevation. Derive an equation for finding the superelevation. A radius of 250m has to be provided at a locality due to site restrictions in a national highway with design speed 100 kmph. Design the superelevation. Should there be restriction in speed?

(20)

ALIAH UNIVERSITY
End-Term Examination- 5th Semester
Session-2021-2022

Year: 3rd Year
Semester: 5th Semester
Subject: Soil Mechanics
Subject Code: CEUGPC14

Full Marks: 80
Time: 3hr.

Answer question no.1 and any three from the rest:

1. Answer the question briefly

[2*10=20]

- i) What is void ratio and degree of saturation?
- ii) What is plasticity index and liquidity index?
- iii) Define superficial velocity and seepage velocity?
- iv) What are the main differences of compaction and consolidation?
- v) Define the terms relative density and activity.
- vi) Write the definition of residual soil and transported soil.
- vii) What is coefficient of curvature?
- viii) What are the main mineralogical groups of soil?
- ix) Why deflocculating agent is used in hydrometer test?
- x) Write the significance of soil classification?

2. a) Give the grain size ranges of different soil types according to IS:1498(1970) (Indian Standard Soil Classification System). [3]

b) Define coarse grained and fine grained soil according to Indian standard Soil Classification System. [2]

c) Define Atterberg's consistency limits with illustration. [5]

d) The weight of a marine clay sample is 50gm. After oven drying the weight of the sample is 30.5gm. Before drying the sample is immersed in mercury and its volume found to be 32cc. Calculate the water content, void ratio and degree of saturation of the sample. Assume $G = 2.68$. [10]

3. a) Explain Darcy's law of fluid flow through porous medium. Show that seepage velocity is always greater than superficial velocity or discharge velocity. [5+3]

c) What is effective stress? Explain. [2]

d) Write the importance of permeability in soil mechanics. [2]

e) An undisturbed sample of clayey soil is found to have a wet weight of 285 N, a dry weight of 250 N, and a total volume of $14 \times 10^3 \text{ cm}^3$ if the specific gravity of soil solids is 2.70, determine the water content, void ratio and the degree of saturation. [8]

4. i) Explain 'Bulking of sand' and define 'Zero Air Void Density'.

ii) With schematic diagram show different type of soil water.

iii) Write the difference between compaction and consolidation and mention the name of the factors affecting compaction.

iv) Illustrate the Plasticity Chart as per Indian Standard Soil classification System and define 'A' line from the chart. [4×5=20]

5. a) What are different soil classification systems?

b) Write the group symbols consisting of a prefix and suffix for various groups as per USCS.

c) What do you mean by effective stress?

d) What is 'Quick Sand'? And discuss the hydraulic condition when Quick Sand condition arrived.

e) Discusses the factors which affect permeability. [5+5+2+4+4=20]

“UG” REGULAR EXAMINATION – Dec. 2021

3RD YEAR (5TH SEMESTER) CIVIL ENGINEERING

STRENGTH OF MATERIALS

MENUGOE03

Full Marks :80

Time – 3 Hours.

Question No. 1 is Compulsory. Answer any five from the rest.

1. Choose the correct alternatives.

[5 X 2=10]

(i) Which of these is a Non-Hookean material ?

- (a) Steel
- (b) Aluminium
- (c) Copper
- (d) Rubber

(ii) A beam which extends beyond its supports can be termed as –

- (a) Over hang beam
- (b) Over span beam
- (c) Tee beam
- (d) Isolated beam

(iii) What are the units of torsional rigidity?

- (a) N-mm^2
- (b) N-mm
- (c) N/mm
- (d) N-mm^4

(iv) For Uniformly Distributed Load over a cantilever beam, the SF Diagram is-

- (a) straight line
- (b) parabolic
- (c) cubic
- (d) None of the above

(v) ----- of column mainly depends upon end conditions.

- (a) Radius of gyration
- (b) Slenderness ratio
- (c) Effective Length
- (d) None of these

2. Deduce the expression of **Euler's Critical** load for the column whose both ends are hinged . [14]

3. An Aluminium rod **ABC** consists of two parts **AB** and **BC** of diameters **40 mm** and **60 mm** with lengths of **300 mm** and **450 mm** respectively. It is subjected to a compressive load of **125 kN** as shown in **Fig.1**. This bar is to be replaced by a steel rod of length **750 mm** to carry the same compressive load. Find the diameter of steel rod if its decrease in length shall not exceed that of the Aluminium rod and the stress in the rod does not exceed **160 N/mm²**. [14]

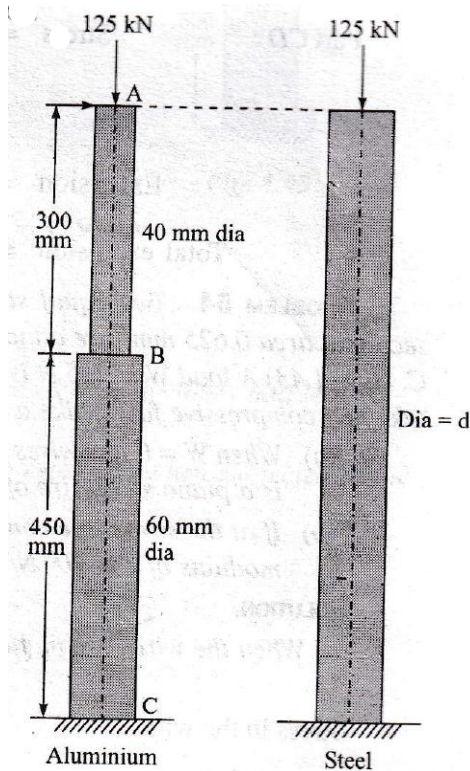


Fig. 1

(2)

4. Three steel plates are welded to form a column section shown in **Fig. 2**. If the effective length of the column is **7.5 mm** , find the factor of safety with respect to buckling for a centric load of **12 kN**. Take, $E = 2 \times 10^5 \text{ N/mm}^2$. [14]

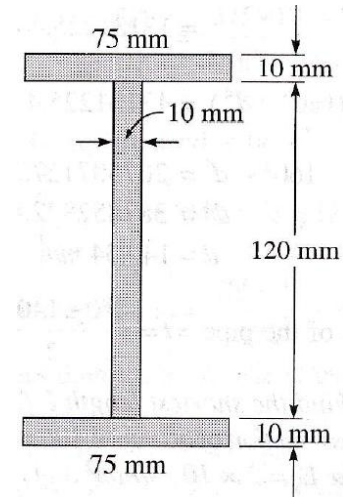


Fig. 2

5. Determine the torque that can be applied to a solid shaft of **20 mm** diameter without exceeding an allowable bending stress of **65 N/mm²**. What torque can be applied if this solid shaft is replaced by a hollow shaft of the same sectional area with the inner diameter equal to half of its outer diameter ? [14]

6. A cantilever beam of length **3m** carries a point load of **60 kN** at a distance of **2m** from the fixed end. If $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 10^8 \text{ mm}^4$, find the slope and the deflection at free end. [14]

7. Draw shear force and bending moment diagram for the overhang beam as shown in **Fig. 3**. [14]

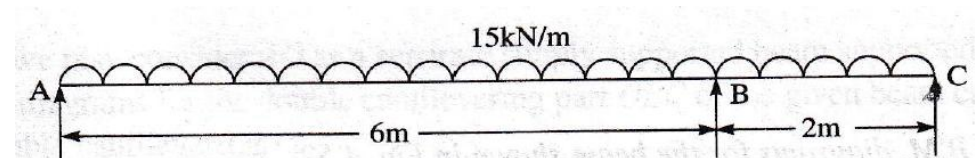


Fig. 3

*****THE END*****